Shivashant Manohar

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Education

VIT–AP University 2023 – 2027
Amaravati, Andhra Pradesh
B.Tech in Computer Science — Artificial Intelligence & Machine Learning **CGPA: 8.05/10**

Technical Skills

Languages: Python, C++, SQL
ML / Deep Learning: PyTorch, Scikit-learn, Transformers, Hugging Face, NumPy, Pandas
Audio / Retrieval: Music Information Retrieval, librosa, Essentia, FAISS, metric learning, sequence modeling
Engineering / Tools: Git, Linux/Unix, Gradio, Matplotlib, Google Cloud Platform

Projects

CURRENTLY WORKING ON

Structure-Aware Hybrid Retrieval for Cover Song Identification 2026 – Present

- Building a cover-song retrieval pipeline on Da-TACOS using chroma/HPCP features, 12-key transposition handling, cross-similarity, and DTW/local alignment as the classical retrieval baseline.
- Implementing dataset loading, WID/PID-aware evaluation, candidate ranking, and retrieval metrics including MAP, MRR, Recall@1, Recall@10, and Recall@100.
- Developing a PyTorch TCN sequence encoder with triplet-loss embeddings and exact FAISS retrieval to learn track-level representations for cover identification.
- Designing controlled ablations, robustness tests, and hybrid reranking; comparing retrieval quality with failure analysis and work-level bootstrap confidence intervals.

Audio Explorer — Interactive Audio AI Preprocessing 2026

- Built an interactive Streamlit application for uploading WAV, MP3, and FLAC audio and inspecting waveform, signal-health, and time–frequency representations.
- Implemented a modular Python audio-processing pipeline using Librosa, NumPy, SciPy, and Plotly, separating feature computation from interactive visualization.
- Developed visualizations for STFT spectrograms, 128-band mel spectrograms, MFCCs with delta features, and beat tracking with tempo and beat-timestamp analysis.

Modern Mini LLM — Decoder-Only Transformer from Scratch 2024

- Implemented a complete PyTorch LLM training pipeline with a byte-level BPE tokenizer, offline caching, memmap-based TinyStories loading, and custom RMSNorm, RoPE, Grouped-Query Attention, and SwiGLU components.
- Built and validated the architecture from first principles, testing individual components before consulting reference implementations.
- Added CLI inference and an interactive Gradio interface, creating an end-to-end path from tokenization and training to usable inference.

Federated Intrusion Detection System — Privacy-Preserving ML 2024

- Designed a federated anomaly-detection pipeline for decentralized healthcare networks, training models without centralizing sensitive client data.
- Implemented lightweight secure aggregation for distributed clients and studied robustness of anomaly detection under decentralized training.
- Evaluated the tradeoff between aggregation efficiency and model accuracy under non-IID client distributions.

Relevant Coursework

Data Structures & Algorithms · Machine Learning · Deep Learning · Artificial Intelligence
Database Management Systems · Probability & Statistics · Computer Networks

Leadership & Community

Google Developer Groups VIT-AP — Cloud & DevOps Team Aug 2024 – Present

- Worked with GCP and cloud-native tooling and contributed to developer workshops and community sessions.